DentiField: Implicit Surface Semantic Field for Intraoral Mesh Segmentation

Supplementary Material

Overview

This supplementary material provides additional details and analyses supporting the main paper. Section A describes the dataset preprocessing pipeline, surface-local query supervision, and the architecture and optimization of DENTIFIELD. Section B presents the implementation and full-surface output construction of the comparison methods and validates our reproductions using released checkpoints. Section C provides detailed definitions of the evaluation metrics. Section D presents additional quantitative and qualitative results, followed by a discussion of failure cases and future work.

A. Implementation Details

In this section, we describe the implementation details of DENTIFIELD. Sec. A.1 specifies the Teeth3DS+ split and point cloud preprocessing, and Sec. A.2 describes the construction of surface-local training queries and their targets. Sec. A.3 details the architecture and pretrained initialization of DENTIFIELD, while Sec. A.4 gives its optimization settings.

A.1. Dataset and Preprocessing

In this subsection, we describe the common Teeth3DS+ [1] split and the preprocessing used to construct the point cloud inputs.

Dataset Split. We use the official Teeth3DS+ partition, containing 1,200 training scans and 600 independent test scans. We form five deterministic, patient-grouped folds from the training split using seed 42. The upper and lower jaws of each patient are assigned to the same fold. Fold0 is used as the validation set in all experiments, resulting in 960 training and 240 validation scans with equal numbers of upper- and lower-jaw scans. The 600 test data are not used during model training.

Preprocessing for Input Point Clouds. Following CrossTooth [14], every point-based and implicit experiment receives a deterministic point cloud of 16K points from each mesh. Following the CrossTooth implementation, we use selective QEM for point cloud sampling. Let $\bar{\mathbf{p}}$ be the mean of the selected vertex coordinates and $s = \max_i \|\mathbf{p}_i - \bar{\mathbf{p}}\|_2$. The normalized coordinate and model input are

$$\tilde{\mathbf{p}}_i = (\mathbf{p}_i - \bar{\mathbf{p}})/s, \quad \mathbf{o}_i = [\tilde{\mathbf{p}}_i,\ \mathbf{n}_i] \in \mathbb{R}^6, \tag{S1}$$

where $\mathbf{n}_i$ is the source vertex normal.

A.2. Surface-Local Query Supervision

Here, we describe how near-surface queries and their semantic and boundary targets are constructed for training. At each optimization step, DENTIFIELD uses 8,192 near-surface queries sampled independently of the 16K point cloud used by the encoder. We obtain each query from an area-weighted sample on the raw triangular surface, apply isotropic Gaussian perturbation with a standard deviation of 0.3 mm, and normalize the resulting coordinate using the same center and scale as the encoder input.

The semantic target is determined from the dominant barycentric vertex at the closest surface point. The auxiliary boundary target is also computed from this projected surface point, ensuring that both targets remain aligned with the annotated mesh despite the off-surface perturbation. Near-surface queries are used only during training. At evaluation, the semantic field is queried at the input point locations for point cloud metrics and at every original mesh vertex for full-surface metrics.

A.3. DentiField Architecture

In this subsection, we describe the detailed architecture design and initialization sources of our DENTIFIELD (see Tab. S1). The surface encoder converts the 16K input point cloud into 4,096 geometry tokens, which are compressed to 64-dimensional posterior modes and projected to 1,024-dimensional semantic latent features. The coordinate-query decoder attends to these features to produce a semantic feature for each query, and the task-specific heads predict tooth classes and the auxiliary boundary target. The table also distinguishes modules initialized from TripoSG [6] from newly initialized or parameter-free components.

Layer name	Input	Output	From TripoSG
Surface Geometry Encoder			
Coordinate and normal features	$\mathbf{X} \in \mathbb{R}^{N\times 6}$	$[\gamma(\mathbf{x}_n), \mathbf{n}_n] \in \mathbb{R}^{N\times 54}$	X
Point projection $\mathrm{MLP}_{\mathrm{in}}$	$N \times 54$	$N \times 512$	O
Token sampler (FPS)	$N \times 512$	$M \times 512$, $M = 4{,}096$	X
Cross-attention	$M \times 512$, $N \times 512$	$M \times 512$	O
Self-attention blocks $\times 8$	$M \times 512$	$M \times 512$	O
Latent Bottleneck and Semantic Latent Decoder			
Posterior projection $\mathrm{MLP}_{\mathrm{enc}}$	$M \times 512$	Posterior parameters in $\mathbb{R}^{M\times 128}$	O
Posterior mode	$M \times 128$	$\mathbf{Z}_{\mathrm{geo}} \in \mathbb{R}^{M\times 64}$	X
Latent projection $\mathrm{MLP}_{\mathrm{dec}}$	$\mathbf{Z}_{\mathrm{geo}} \in \mathbb{R}^{M\times 64}$	$M \times 1024$	O
Latent self-attention blocks $\times 16$	$M \times 1024$	$M \times 1024$	O
Coordinate-query Semantic Decoder			
Query frequency embedding	$\mathbf{Q} \in \mathbb{R}^{Q\times 3}$	$Q \times 51$	X
Query cross-attention	$Q \times 51$, $M \times 1024$	$\mathbf{H} \in \mathbb{R}^{Q\times 1024}$	O
Task-specific Prediction Heads			
Semantic MLP (5 layers)	$\mathbf{H} \in \mathbb{R}^{Q\times 1024}$	$\mathbf{O}^{\mathrm{sem}} \in \mathbb{R}^{Q\times 17}$	X
Boundary MLP (2 layers)	$\mathbf{H} \in \mathbb{R}^{Q\times 1024}$	$\mathbf{O}^{\mathrm{bnd}} \in \mathbb{R}^{Q\times 2}$	X

Table S1. **Layer-level architectural details of DENTIFIELD.** The 'From TripoSG' indicates whether a module is initialized from pretrained TripoSG weights (O); X denotes a newly initialized module or a parameter-free operation. All TripoSG-initialized components are jointly fine-tuned. The semantic and auxiliary boundary heads are initialized from scratch, and the boundary head is used only during training.

Method	Reference source code
TSegNet [5]	https://github.com/snake-head/tsegnet
ToothGroupNetwork [7]	https://github.com/limhoyeon/ToothGroupNetwork
Point Transformer V3 [13]	https://github.com/Pointcept/PointTransformerV3
DITTO [12]	https://github.com/vision3d-lab/ditto
3DTeethSAM [9]	https://github.com/Crisitofy/3DTeethSAM
CrossTooth [14]	https://github.com/XiShuFan/CrossTooth_CVPR2025
PMTSeg [15]	https://github.com/znshje/PMTSeg

Table S2. Reference source code for compared comparison methods.

A.4. Additional Implementation Details

We describe the optimization settings of the final DENTIFIELD configuration. We train DENTIFIELD with cross-entropy and Lovász-Softmax [2] losses, with unit weight for the Lovász term, and an auxiliary binary boundary cross-entropy loss with a weight of 0.3. We use AdamW [8] with weight decay 0.01 and learning rates of 10^{-4} for the newly initialized heads, 2×10^{-5} for the decoder, and 10^{-5} for the encoder. Training uses BF16 mixed precision, gradient clipping at 1.0, a two-epoch warm-up, and a cosine-decay learning-rate schedule for 450 epochs. The training takes approximately 18 hours on an RTX 3090 GPU. We train most comparison methods for 450 epochs for full convergence and fair comparison. Most methods converge within 200 epochs.

B. Baseline Details

Sec. B.1 lists the baseline source code and training protocols, and Sec. B.2 explains how each baseline produces a full-surface segmentation. Sec. B.3 validates the reproduced baseline results and evaluation pipeline using released checkpoints.

B.1. Comparison Methods

For fair comparison, we have implemented and reproduced most comparison methods using their official source code. The referenced source codes are listed in Tab. S2. For TSegNet [5], since the original authors have not released an official code repository, the listed TSegNet repository is an unofficial reimplementation. However, because the publicly available TSegNet [5] reimplementation has reproducibility issues, we use the pretrained checkpoint released with ToothGroupNetwork [7] instead of retraining TSegNet. Following their respective training schedules, we train ToothGroupNetwork [7] for 60 epochs

and Point Transformer V3 [13] for 450 epochs.

As an implicit baseline, we train DITTO [12] by modifying the input dimension to six channels so that the model receives both point coordinates and surface normals. As an image-based baseline, we reimplement 3DTeethSAM [9]. We render the complete mesh from seven predefined views and transfer the masks back to the mesh through visibility- and depth-aware voting. We train its localization stage for 30 epochs, initialize the joint stage from the resulting checkpoint, and train the joint stage for 70 additional epochs (epochs 31-100).

For CrossTooth [14] and PMTSeg [15], we train both methods based on the official code for 450 epochs. However, PMTSeg requires real-world intraoral images used in original study of PMTSeg, that cannot be acquired, we provide the mesh rendeded images instead of the dataset.

B.2. Details of Full-Surface Output Construction for Comparison Methods

To compute mesh-level metrics, we construct a full-resolution segmentation map on the original scan mesh for each comparison method. Because the baselines produce different output representations, we use the following method-specific construction paths.

Point-based methods. For TSegNet [5], we predict semantic labels using its 24K points based on its inference code, whereas ToothGroupNetwork [7], Point Transformer V3 [13], CrossTooth [14], and PMTSeg [15] predict labels at the canonical 16K sampled point locations. For each method, we anchor these predictions to their corresponding source mesh vertices and assign every remaining vertex the label of its nearest anchor under shortest-path distance on the triangle-edge graph. Although PMTSeg uses synthetic mesh renders as auxiliary inputs, its evaluated 3D point cloud output remains a point-based prediction and therefore follows the same propagation procedure.

Point-to-surface transfer. Point-based methods predict semantic labels only at sampled point locations, and their released implementations use different procedures to transfer these labels to the native mesh. For a controlled comparison, the main experiment applies the same nearest-anchor propagation based on shortest-path distances on the triangle-edge graph to every point-based baseline. We denote this shared protocol as `Graph geodesic`. This protocol preserves the sampled point predictions while standardizing the construction of full-surface outputs across methods. The transfer is not applied to DENTIFIELD, which directly evaluates its implicit semantic field at every native mesh vertex.

For completeness, Tab. S3 additionally evaluates all point-based baselines using the Euclidean three-nearest-neighbor transfer provided by the official implementation of CrossTooth [14], denoted as `CrossTooth 3-NN`. Because the sampled point predictions are held fixed, the point cloud metrics remain identical between the two transfer protocols, while only the full-surface boundary metrics can change. Across all point-based baselines, `CrossTooth 3-NN` yields consistently lower Boundary Dice and F1 than `Graph geodesic`. The shared graph-geodesic protocol therefore provides the point-based baselines with the stronger of the two transfer procedures. Nevertheless, DENTIFIELD achieves the best boundary performance without point-to-surface propagation, making the main comparison conservative with respect to DENTIFIELD.

Comparison Methods	Point-to-surface	OA↑	T-mIoU↑	mIoU↑	DSC↑	B-IoU↑	B-Dice@0.2↑	B-Dice@0.1↑	B-F1@0.2↑	B-F1@0.1↑
DITTO [12]	-	92.40 ± 7.98	84.12 ± 13.13	82.36 ± 14.68	87.62 ± 14.30	73.07 ± 14.41	74.47 ± 17.35	50.40 ± 14.68	77.68 ± 16.70	53.36 ± 14.61
3DTeethSAM [9]	-	92.52 ± 7.85	85.45 ± 11.63	84.25 ± 13.41	88.96 ± 13.53	75.71 ± 12.60	81.84 ± 13.59	56.57 ± 11.60	84.54 ± 13.47	59.90 ± 11.98
TSegNet [5]	Graph geodesic	83.87 ± 11.05	66.39 ± 20.06	65.61 ± 20.50	72.65 ± 20.87	54.32 ± 18.09	54.64 ± 19.65	32.04 ± 13.14	57.88 ± 20.28	34.27 ± 13.74
	CrossTooth 3-NN	83.87 ± 11.05	66.39 ± 20.06	65.61 ± 20.50	72.65 ± 20.87	54.32 ± 18.09	39.23 ± 14.02	20.61 ± 8.32	42.79 ± 15.01	22.62 ± 8.82
ToothGroupNetwork [7]	Graph geodesic	89.94 ± 11.21	77.47 ± 21.94	76.66 ± 22.60	80.45 ± 22.82	70.27 ± 21.26	66.04 ± 19.43	42.43 ± 14.45	69.43 ± 20.28	45.86 ± 15.43
	CrossTooth 3-NN	89.94 ± 11.21	77.47 ± 21.94	76.66 ± 22.60	80.45 ± 22.82	70.27 ± 21.26	60.18 ± 18.19	35.81 ± 12.75	63.84 ± 19.17	39.02 ± 13.66
Point Transformer V3 [13]	Graph geodesic	92.10 ± 7.98	82.84 ± 13.36	80.97 ± 14.97	86.43 ± 14.75	71.73 ± 14.21	65.38 ± 13.71	38.89 ± 9.87	70.34 ± 14.05	42.83 ± 10.30
	CrossTooth 3-NN	92.10 ± 7.98	82.84 ± 13.36	80.97 ± 14.97	86.43 ± 14.75	71.73 ± 14.21	58.72 ± 12.42	32.22 ± 8.47	63.48 ± 12.88	35.49 ± 8.78
CrossTooth [14]	Graph geodesic	94.20 ± 7.27	88.06 ± 11.32	86.97 ± 12.88	90.94 ± 12.95	79.77 ± 12.31	73.60 ± 11.28	46.43 ± 9.04	78.04 ± 11.52	50.78 ± 9.46
	CrossTooth 3-NN	94.20 ± 7.27	88.06 ± 11.32	86.97 ± 12.88	90.94 ± 12.95	79.77 ± 12.31	67.43 ± 10.55	39.31 ± 8.23	72.07 ± 10.92	43.22 ± 8.61
PMTSeg† [15]	Graph geodesic	94.08 ± 7.34	87.65 ± 11.88	86.50 ± 13.51	90.38 ± 13.32	78.84 ± 13.39	72.63 ± 12.69	46.39 ± 10.01	77.08 ± 12.84	50.69 ± 10.37
	CrossTooth 3-NN	94.08 ± 7.34	87.65 ± 11.88	86.50 ± 13.51	90.38 ± 13.32	78.84 ± 13.39	67.35 ± 11.71	39.62 ± 9.06	71.88 ± 12.01	43.44 ± 9.34
DENTIFIELD (ours)	-	$\mathbf{94.44 \pm 6.96}$	$\mathbf{88.47 \pm 10.66}$	$\mathbf{87.70 \pm 12.21}$	$\mathbf{91.63 \pm 12.36}$	$\mathbf{80.67 \pm 11.50}$	$\mathbf{87.65 \pm 12.33}$	$\mathbf{64.94 \pm 10.91}$	$\mathbf{89.63 \pm 12.07}$	$\mathbf{67.18 \pm 10.91}$

Table S3. **Quantitative comparison of point-to-surface transfer protocols.** For each point-based baseline, we compare the `Graph geodesic` point-to-surface transfer used in the main experiment with the `CrossTooth 3-NN` from the official CrossTooth implementation. Consequently, since the sampled point predictions are held fixed, all point cloud metrics are identical within each pair of rows, while full-surface Boundary Dice and Boundary F1 can differ. DITTO and DENTIFIELD directly query every native mesh vertex, whereas 3DTeethSAM produces dense mesh labels using its official multi-view voting, component filtering, and Graph Cut [3] pipeline.

Image-based method. We follow the inference pipeline provided by the released official implementation of 3DTeethSAM [9], using its documented seven-view inference setting. We render the native input mesh from seven predefined viewpoints and predict class-specific two-dimensional tooth masks in each view. The predicted masks are sampled at the projections of visible mesh vertices, and per-vertex labels are determined through visibility- and inverse-depth-weighted voting across views. The method then retains the largest connected component for each tooth class and applies Graph Cut refinement to obtain dense native-mesh labels.

Implicit method. We adapt DITTO [12] from occupancy prediction to semantic segmentation by modifying its input and output heads to receive point coordinates and surface normals and predict 17 semantic classes. After this adaptation, DITTO uses the same input and output format as DENTIFIELD. We therefore evaluate DITTO using the same full-surface inference procedure, directly querying every native mesh vertex without point-to-surface propagation.

B.3. Validation with Released Baseline Checkpoints

To assess the reliability of our reproduced baseline results, we additionally evaluate the released checkpoints of TSegNet [5], ToothGroupNetwork [7], CrossTooth [14], and 3DTeethSAM [9] on the same 600-scan test set used in the main experiments (see Tab. S4). Because these checkpoints are independent of our baseline retraining, this evaluation reduces the concern that the main comparison is affected by weak baseline optimization or checkpoint selection. The results preserve the main conclusion, with DENTIFIELD outperforming all released baseline checkpoints across all nine metrics.

Method	OA↑	T-mIoU↑	mIoU↑	DSC↑	B-IoU↑	B-Dice@0.1↑	B-Dice@0.2↑	B-F1@0.1↑	B-F1@0.2↑
TSegNet [5]	83.90 ± 11.04	66.44 ± 20.05	65.66 ± 20.48	72.69 ± 20.86	54.37 ± 18.08	32.08 ± 13.13	54.70 ± 19.62	34.31 ± 13.72	57.94 ± 20.24
ToothGroupNetwork FPS [7]	82.28 ± 19.07	62.97 ± 33.43	62.62 ± 33.49	67.45 ± 35.72	54.63 ± 29.38	36.82 ± 20.56	59.95 ± 32.18	38.75 ± 21.55	62.21 ± 33.29
CrossTooth [14]	87.94 ± 8.21	76.16 ± 13.30	74.90 ± 14.07	82.26 ± 13.91	62.84 ± 12.48	27.71 ± 6.27	50.69 ± 10.64	29.11 ± 6.37	53.32 ± 10.53
3DTeethSAM [9]	$\underline{93.33 \pm 7.36}$	$\underline{86.65 \pm 10.94}$	$\underline{85.68 \pm 12.54}$	$\underline{90.35 \pm 12.68}$	$\underline{77.17 \pm 11.79}$	$\underline{57.63 \pm 10.89}$	$\underline{83.39 \pm 12.67}$	$\underline{60.97 \pm 11.23}$	$\underline{86.07 \pm 12.48}$
DENTIFIELD (ours)	$\mathbf{94.44 \pm 6.96}$	$\mathbf{88.47 \pm 10.66}$	$\mathbf{87.70 \pm 12.21}$	$\mathbf{91.63 \pm 12.36}$	$\mathbf{80.67 \pm 11.50}$	$\mathbf{64.94 \pm 10.91}$	$\mathbf{87.65 \pm 12.33}$	$\mathbf{67.18 \pm 10.91}$	$\mathbf{89.63 \pm 12.07}$

Table S4. **Validation with released baseline checkpoints.** We evaluate the comparison methods with our 600-scan test dataset using the provided pretrained baseline checkpoints. To Mean $\pm$ standard deviation is reported over the same 600-scan test set used in the main experiments. The released checkpoints may reflect method-specific training, data-partition, and model-selection protocols; therefore, this table complements rather than replaces the controlled fixed-split comparison. Bold and underlined values indicate the best and second-best mean, respectively.

C. Details of Evaluation Metrics

C.1. Point Cloud Evaluation Metrics

Following the evaluation protocols of 3DTeethSAM [9] and CrossTooth [14], we report OA, T-mIoU, mIoU, DSC, and BoundaryIoU on the point cloud. Let $\mathbf{p}_i$ be the physical coordinate of evaluation point i, and let y_i and $\hat{y}_i$ denote its ground-truth and predicted labels. Let $P_c = \{i \mid \hat{y}_i = c\}$ and $G_c = \{i \mid y_i = c\}$ for tooth class $c \in \{1, \ldots, 16\}$. All reported metrics are computed per scan and averaged over the 600 test scans. Overall accuracy includes gingiva:

$$\mathrm{OA} = \frac{1}{N_{\mathrm{eval}}} \sum_{i=1}^{N_{\mathrm{eval}}} \mathbb{1}[\hat{y}_i = y_i]. \tag{S2}$$

We define $\mathcal{C}_T = \{c \mid |G_c| > 0\}$ and $\mathcal{C}_U = \{c \mid |P_c \cup G_c| > 0\}$. T-mIoU averages only tooth classes present in the ground truth. mIoU and DSC average classes present in either the prediction or ground truth, assigning zero overlap to a spurious predicted class:

$$\text{T-mIoU} = \frac{1}{|\mathcal{C}_T|} \sum_{c \in \mathcal{C}_T} \frac{|P_c \cap G_c|}{|P_c \cup G_c|}. \tag{S3}$$

$$\mathrm{mIoU} = \frac{1}{|\mathcal{C}_U|} \sum_{c \in \mathcal{C}_U} \frac{|P_c \cap G_c|}{|P_c \cup G_c|}. \tag{S4}$$

$$\mathrm{DSC} = \frac{1}{|\mathcal{C}_U|} \sum_{c \in \mathcal{C}_U} \frac{2|P_c \cap G_c|}{|P_c| + |G_c|}. \tag{S5}$$

All class means exclude background.

BoundaryIoU is computed on the point cloud after restoring physical millimeter coordinates. For a radius $r = 2$ mm, its class-aware boundary region is

$$B_c(y; r) = \{i \mid y_i = c,\ \exists j : \|\mathbf{p}_i - \mathbf{p}_j\|_2 \le r,\ y_j \neq y_i\}. \tag{S6}$$

BoundaryIoU is the class mean of the IoU between $B_c(\hat{y}; r)$ and $B_c(y; r)$ for classes with a nonempty union.

C.2. Full-Surface Boundary Evaluation

Boundary Dice and Boundary F1 are motivated by surface-distance-based boundary evaluation [4, 10]. Boundary Dice is evaluated on the full-resolution output mesh. For each tooth class c, let $\mathcal{B}_c^{\text{pred}}$ and $\mathcal{B}_c^{\text{gt}}$ denote the predicted and ground-truth boundary-vertex index sets. A vertex belongs to either set when it is assigned class c and is incident to a mesh edge whose endpoint labels differ. Each boundary vertex receives a weight equal to the sum of its incident transition-edge lengths, denoted by $w_{c,v}^{\text{pred}}$ or $w_{c,v}^{\text{gt}}$ according to the corresponding segmentation. We define

$$m_\tau(v, \mathcal{S}) = \mathbb{1}\left[\min_{v' \in \mathcal{S}} \|\mathbf{v}_v - \mathbf{v}_{v'}\|_2 \le \tau\right]. \tag{S7}$$

We set $m_\tau(v, \emptyset) = 0$ and use a tolerance of $\tau = 0.1$ and $\tau = 0.2$ mm. The class score is

$$\mathrm{BD}_c^\tau = \frac{\sum_{v \in \mathcal{B}_c^{\text{pred}}} w_{c,v}^{\text{pred}} m_\tau(v, \mathcal{B}_c^{\text{gt}}) + \sum_{v \in \mathcal{B}_c^{\text{gt}}} w_{c,v}^{\text{gt}} m_\tau(v, \mathcal{B}_c^{\text{pred}})}{\sum_{v \in \mathcal{B}_c^{\text{pred}}} w_{c,v}^{\text{pred}} + \sum_{v \in \mathcal{B}_c^{\text{gt}}} w_{c,v}^{\text{gt}}}. \tag{S8}$$

Boundary Dice is the mean of Eq. (S8) over tooth classes with a predicted or target boundary. We additionally report Boundary F1 at 0.1 and 0.2 mm, which gives each matched boundary vertex equal weight, alongside the corresponding edge-length-weighted Boundary Dice scores.

D. Additional Results and Analysis

D.1. Uncommon-Sample Evaluation by Tooth Count

Table S5 decomposes the aggregate uncommon-sample comparison in Tab. 3 of the main paper by the number of present teeth, revealing whether the overall trend is consistent across tooth-count configurations. Across every reported group, DENTIFIELD achieves the highest B-IoU, as well as the highest B-Dice and B-F1 scores at both tolerances. It also ranks first across all metrics for samples with 10, 11, and 16 teeth, and in the 15-tooth group only its T-mIoU is slightly below that of CrossTooth [14]. For the nine-tooth group, DENTIFIELD retains the strongest boundary scores, although CrossTooth [14] obtains a slightly higher OA and PMTSeg [15] a slightly higher DSC. The boundary advantage observed across multiple groups supports the aggregate uncommon-sample result reported in the main paper, rather than attributing the gain to a single tooth-count configuration. Because the nine-tooth group contains only three test scans, its per-group ranking should be interpreted as descriptive rather than conclusive.

D.2. Inference Cost

Table S6 reports the per-scan inference cost of DENTIFIELD and the comparison methods that produce labels on the complete mesh. All methods are measured on the same 600 test scans using a single RTX 4090 GPU, with one scan processed at a time. The reported time excludes disk I/O and input point sampling, and the propagation of CrossTooth labels to mesh vertices runs on the CPU (Intel Core Ultra 7 265K). DENTIFIELD encodes each scan once and evaluates all mesh vertices in chunks of 8,192 queries; for DITTO, we likewise encode each scan once and decode its vertices in chunks of the same size. DENTIFIELD has the lowest end-to-end latency because it requires neither label transfer nor multi-view rendering. Its peak memory grows linearly with the number of mesh vertices and remains below that of 3DTeethSAM, but it is higher than that of the lighter point-based and implicit baselines.

D.3. Additional Qualitative Comparisons

Figure S3 extends the qualitative comparison in the main paper to four additional test scans. For each scan, we show the semantic prediction and the corresponding boundary overlay on the complete output mesh. As observed in the main paper, competing pipelines can merge adjacent teeth or assign local surface regions to an incorrect tooth class, particularly near

#Teeth	Method	N	OA↑	T-mIoU↑	mIoU↑	DSC↑	B-IoU↑	B-Dice@0.2↑	B-Dice@0.1↑	B-F1@0.2↑	B-F1@0.1↑
9	TSegNet [5]	3	74.05	38.33	33.11	38.99	21.85	23.36	12.93	25.22	14.09
	ToothGroupNetwork [7]		68.77	19.80	13.49	14.76	12.56	11.51	7.91	12.08	8.39
	Point Transformer V3 [13]		86.22	65.21	46.74	50.14	38.75	36.60	20.86	39.81	23.31
	DITTO [12]		84.01	58.45	45.95	51.10	35.02	31.06	17.28	34.21	19.21
	3DTeethSAM [9]		86.44	65.32	54.01	57.20	46.50	49.27	28.94	51.65	30.77
	CrossTooth [14]		**88.05**	68.87	61.16	64.35	54.78	51.79	31.51	54.78	34.00
	PMTSeg† [15]		87.88	68.63	63.97	**66.45**	57.09	53.84	33.65	56.78	36.56
	DENTIFIELD (ours)		87.98	**69.51**	**64.03**	66.20	**58.99**	**61.80**	**42.53**	**63.28**	**43.89**
10	TSegNet [5]	12	75.55	58.51	55.45	60.05	46.89	49.47	29.88	51.99	31.74
	ToothGroupNetwork [7]		70.86	44.14	40.99	43.90	37.62	37.28	23.85	39.15	25.57
	Point Transformer V3 [13]		77.96	62.20	52.99	57.79	45.27	42.62	25.63	46.15	28.54
	DITTO [12]		80.05	69.96	62.84	66.40	55.43	58.77	40.05	60.96	42.15
	3DTeethSAM [9]		81.29	71.94	65.74	69.39	58.03	64.98	44.70	66.95	47.27
	CrossTooth [14]		81.54	71.76	67.04	70.36	60.72	58.52	36.82	61.88	40.19
	PMTSeg† [15]		81.63	69.22	61.82	66.13	53.05	50.18	32.60	53.65	35.77
	DENTIFIELD (ours)		**83.16**	**73.84**	**70.95**	**73.91**	**65.05**	**72.19**	**54.86**	**73.83**	**56.50**
11	TSegNet [5]	25	85.62	70.25	66.61	72.81	55.50	55.15	33.36	58.53	35.76
	ToothGroupNetwork [7]		85.07	69.62	66.07	69.40	60.32	56.80	37.20	59.54	39.74
	Point Transformer V3 [13]		87.22	75.47	68.58	73.67	59.16	55.02	33.59	59.27	36.99
	DITTO [12]		88.71	78.61	71.98	77.10	62.41	62.32	42.77	65.79	45.62
	3DTeethSAM [9]		88.42	81.02	75.40	79.35	67.67	73.47	51.68	75.82	54.55
	CrossTooth [14]		89.88	81.43	76.79	80.54	69.76	65.42	42.49	69.44	46.20
	PMTSeg† [15]		88.66	78.99	73.53	77.56	64.96	60.76	39.79	64.81	43.39
	DENTIFIELD (ours)		**90.38**	**83.69**	**79.35**	**82.78**	**72.75**	**78.79**	**59.54**	**80.74**	**61.35**
15	TSegNet [5]	9	70.25	37.89	37.76	45.86	28.68	25.52	11.43	27.62	12.78
	ToothGroupNetwork [7]		83.61	60.50	59.82	64.37	53.81	50.32	26.54	52.48	29.02
	Point Transformer V3 [13]		82.49	61.75	60.74	67.04	53.41	46.45	22.88	51.00	26.21
	DITTO [12]		85.82	69.23	68.37	75.89	58.90	55.48	29.34	58.92	32.24
	3DTeethSAM [9]		85.62	70.71	66.61	72.11	59.33	64.90	38.32	66.66	40.59
	CrossTooth [14]		89.93	**79.66**	78.14	82.42	71.75	63.95	33.84	68.24	38.03
	PMTSeg† [15]		90.26	79.13	78.70	83.05	72.09	63.62	34.25	68.49	38.74
	DENTIFIELD (ours)		**90.47**	79.42	**78.71**	**83.24**	**72.90**	**80.21**	**51.86**	**81.45**	**54.15**
16	TSegNet [5]	22	64.15	31.53	31.53	38.02	25.44	25.42	11.87	27.28	13.11
	ToothGroupNetwork [7]		83.20	60.02	60.02	64.29	53.89	50.20	26.77	52.64	29.37
	Point Transformer V3 [13]		84.87	67.70	67.70	75.40	58.20	47.28	23.44	52.61	27.25
	DITTO [12]		89.68	77.08	77.08	84.56	66.19	63.20	34.72	66.67	38.09
	3DTeethSAM [9]		86.93	75.48	75.48	80.25	68.18	72.80	43.74	74.46	46.26
	CrossTooth [14]		95.36	89.28	89.28	93.71	81.77	71.67	38.68	76.36	43.32
	PMTSeg† [15]		89.06	77.37	77.37	81.66	70.98	61.93	33.70	66.82	38.26
	DENTIFIELD (ours)		**96.18**	**90.78**	**90.78**	**94.93**	**84.00**	**91.35**	**59.27**	**92.49**	**61.81**

Table S5. **Quantitative comparisons on uncommon samples stratified by the number of present teeth.** N denotes the number of test scans in each tooth-count group. Following the uncommon-sample definition in the main paper, we evaluate DENTIFIELD and the baselines on samples with 9-11 or 15-16 present teeth. Bold and underlined values indicate the best and second-best results, respectively.

Method	Render	Network	Transfer	Total	Peak mem.
CrossTooth [14]	–	0.068	0.318	0.386	0.15
DITTO [12]	–	0.263	–	0.263	0.84
3DTeethSAM [9]	0.129	0.654	2.298	3.081	6.71
DENTIFIELD (ours)	–	0.121	–	**0.121**	3.77

Table S6. **Per-scan inference cost on a single RTX 4090 GPU.** Times are in seconds, averaged over the 600 test scans (113K vertices on median, up to 226K). *Transfer* denotes label propagation from points to mesh vertices for CrossTooth, and mask projection, multi-view voting, and Graph Cut refinement for 3DTeethSAM. Peak memory is the maximum allocated GPU memory over the test scans in GiB, including model weights.

narrow inter-tooth regions. In contrast, DENTIFIELD more consistently preserves the separation of adjacent tooth instances and follows the corresponding ground-truth boundaries across the four scans. These additional examples further support the

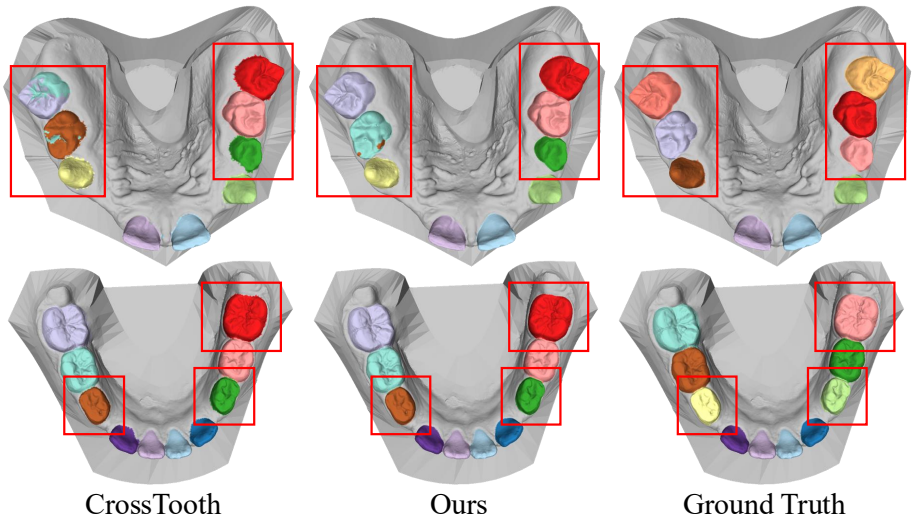


Figure S1. **Failure cases in uncommon samples.** We show two uncommon samples with missing teeth. Both DENTIFIELD and CrossTooth [14] can misclassify remaining tooth classes when the usual inter-tooth spatial cues are absent.

quantitative improvement in complete-surface boundary fidelity reported in the main paper.

D.4. Failure Cases and Future Work

Failure cases. Although DENTIFIELD achieves the strongest segmentation performance, it can still misclassify the tooth class of a remaining tooth in uncommon samples with missing teeth. As shown in Fig. S1, this failure pattern occurs for both DENTIFIELD and baselines. This shared pattern suggests that the training data insufficiently covers variations in tooth count and shape, which can make models rely on absolute tooth positions. Because missing teeth alter the spatial arrangement of the remaining teeth, reliance on absolute tooth positions is unreliable. Therefore, training on Teeth3DS+ [1] alone does not establish robustness to uncommon samples.

Future work. Figure S2 presents a two-dimensional proof of concept in which a vision-language model (VLM) agent using GPT Image 2 [11] removes a target tooth while preserving the surrounding dental structures. This result motivates extending the same agent-guided editing strategy to three-dimensional intraoral scans. We plan to use the VLM agent to specify uncommon tooth configurations and guide geometry-aware generation of the corresponding 3D meshes. The resulting augmented scans will expand the coverage of uncommon dentitions in the training distribution and may reduce reliance on absolute tooth positions.

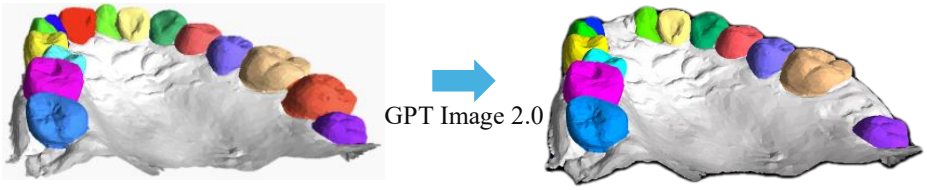


Figure S2. **Two-dimensional proof of concept for VLM-agent-guided uncommon-tooth augmentation.** Given a dental image and a target editing instruction, the VLM agent generates a missing-tooth configuration while preserving the surrounding dental structures.

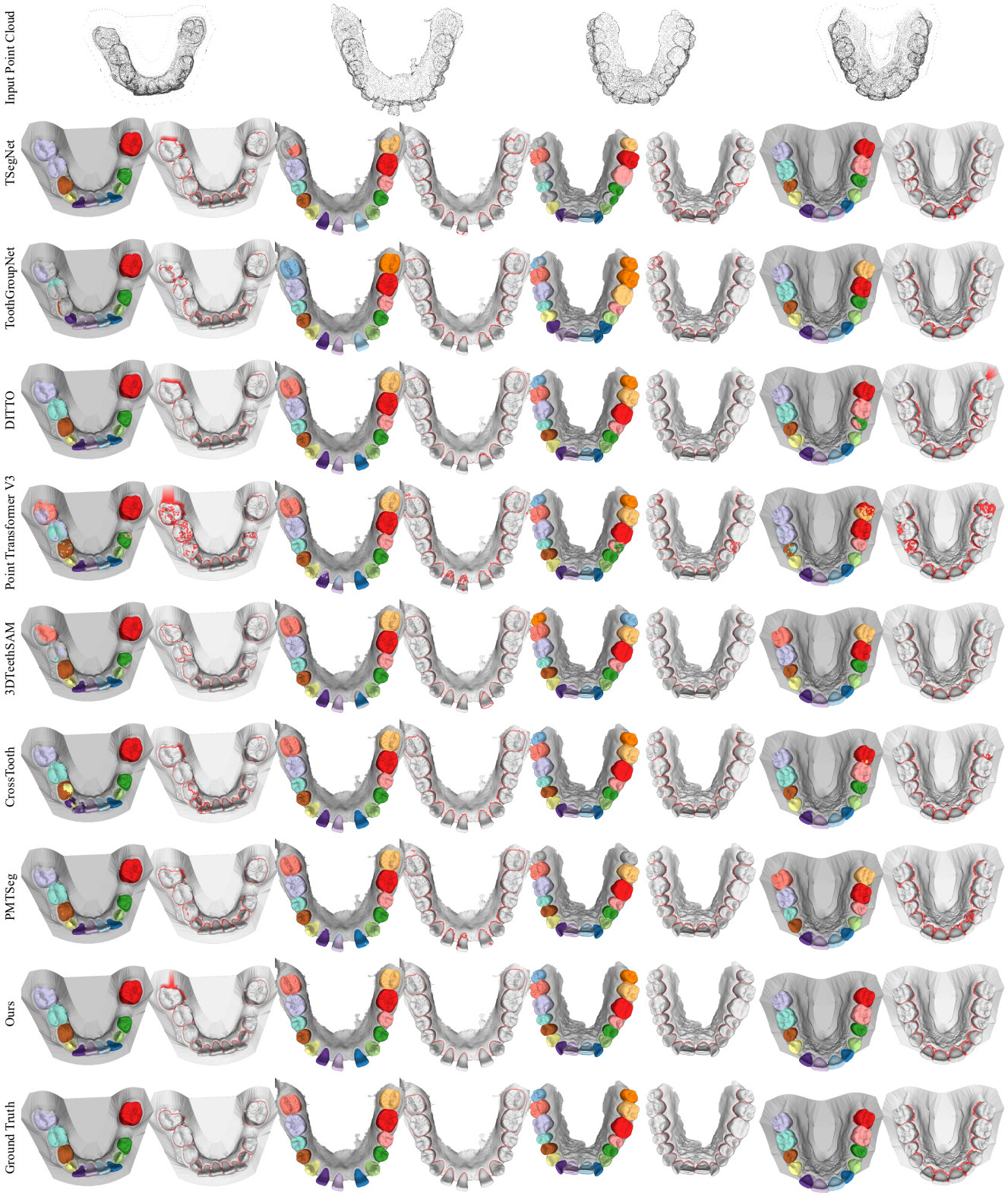


Figure S3. **Additional qualitative comparisons of DENTIFIELD and comparison methods.** DENTIFIELD predicts tooth categories consistent with the ground truth and produces clear inter-tooth and tooth-gingiva boundaries.

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